

EFY Note

The source code of this project is available for free download at source.efymag.com

LCD library is used for display and Wire library is used for I2C communication. Before uploading the code to ATmega328, do not forget to include the related files (Atmega328_expander_library_files.zip) in the Arduino Library.

For uploading the code to ATmega328, use an Arduino Uno board with bootloader. After uploading the code, remove ATmega328 chip from Arduino Uno board and insert it into the circuit or PCB.

Construction and testing

An actual-size PCB layout for the ATmega328 analogue I/O port expander is shown in Fig. 3 and its components layout in Fig. 4. After assembling the circuit on PCB, con-

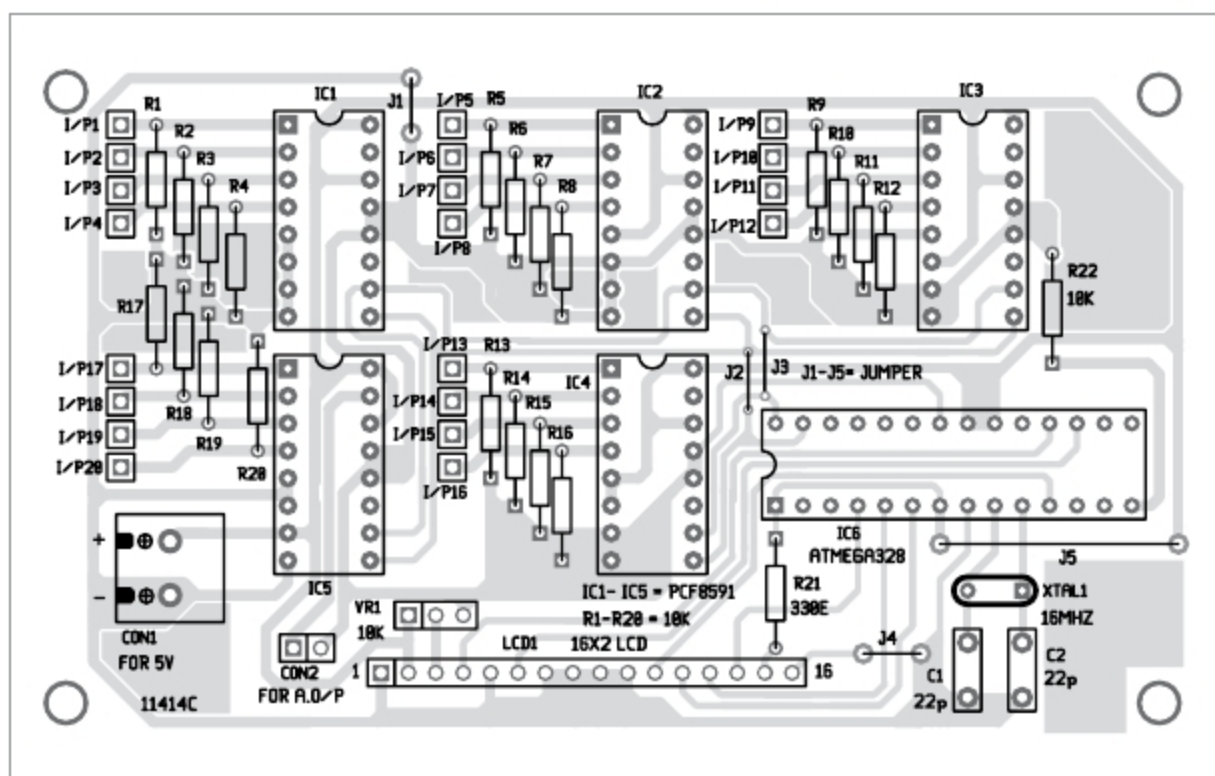


Fig. 4: Components layout of the PCB

nect 5V across CON1.

To test the working of the circuit, just connect a variable resistor or preset between ground and 5V power source with its wiper (middle pin) connected to any of the analogue pins of IC5. Vary the preset and check the output voltage on LCD. You can also

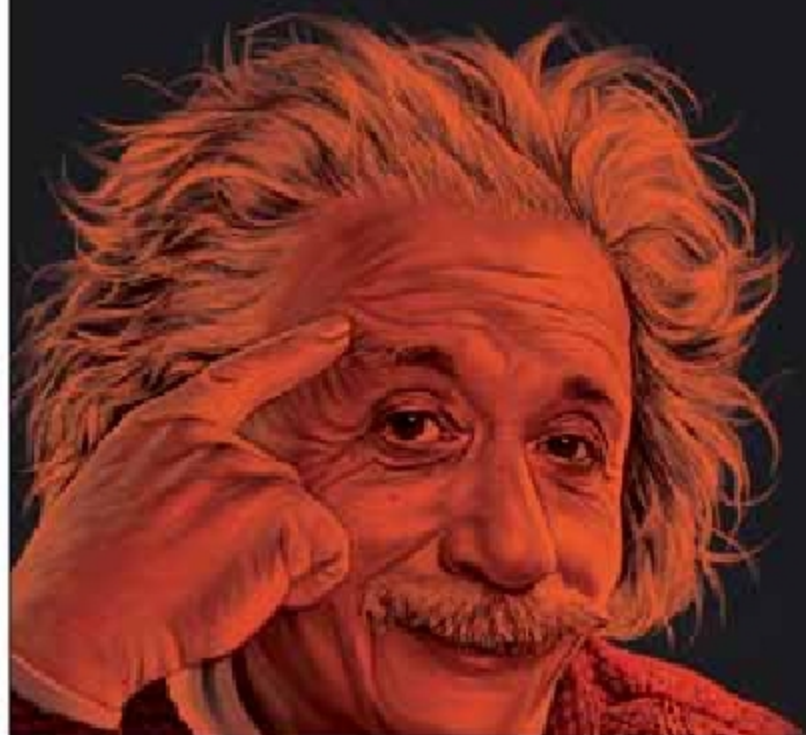
check the output voltage at connector CON2 on a digital voltmeter. **EFY**



S. Basilea Ruban is an electronics and communication engineer, circuit designer, and embedded programmer

Once you stop learning, you start dying.

—Albert Einstein



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